

RangeViT

3D LiDAR Semantic Segmentation with ViTs

Bridging the Gap Between Speed and 3D Accuracy



| The Research Objective



Segmentation

Categorizing every 3D point (Road, Car, Tree, etc.) in outdoor scenes.



Efficiency

Enabling real-time performance on autonomous vehicle hardware.

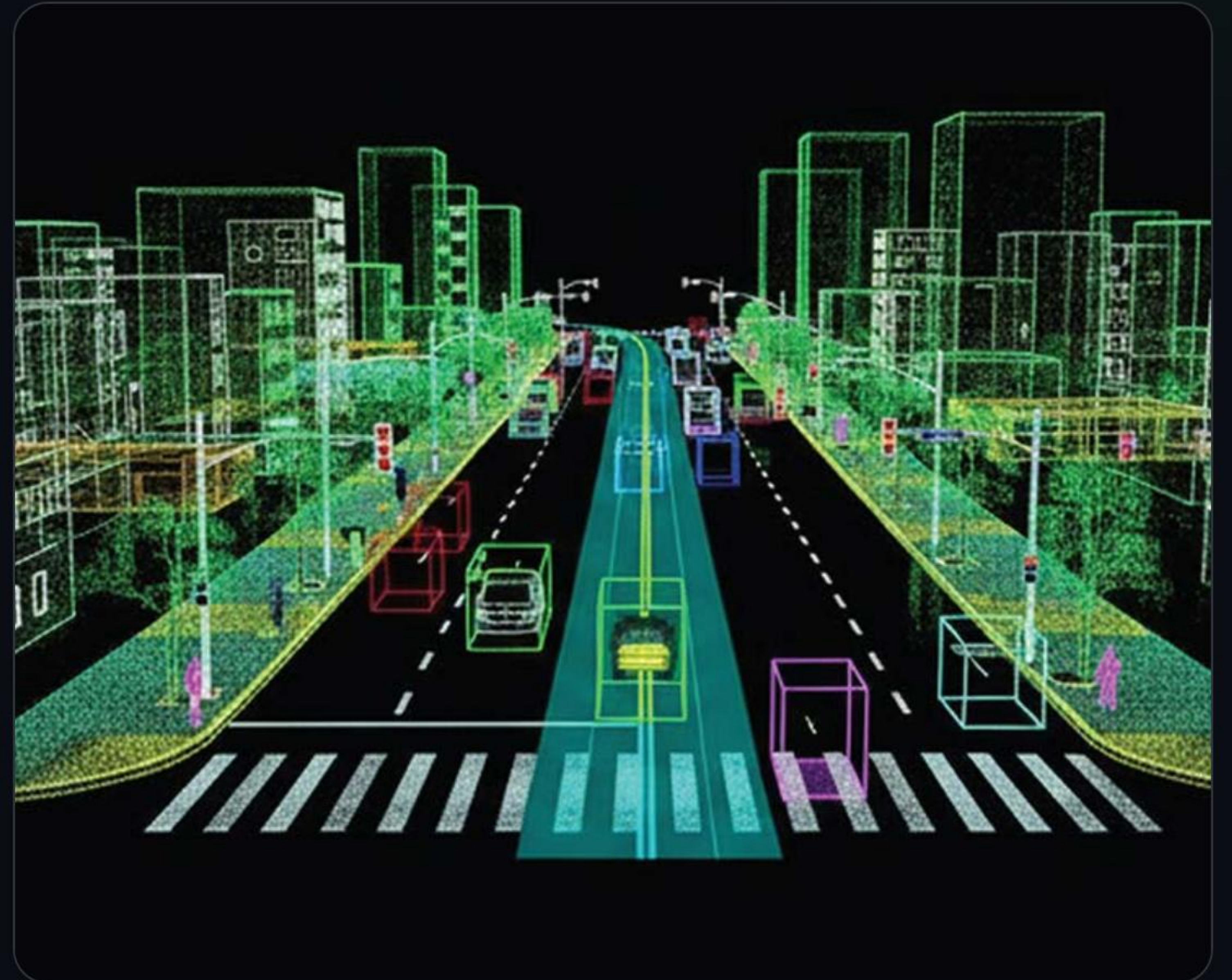


Transfer Power

Using 2D Image "Brains" to solve 3D spatial problems.

LiDAR 101: Understanding the Input

- > **Raw Data:** A "Point Cloud" of (x, y, z, i) coordinates.
- > **Scale:** Outdoor scans capture 100,000+ points per frame.
- > **Challenge:** Data is sparse, unordered, and computationally heavy to process directly in 3D.



Why ViTs are Hard for LiDAR

Computational Explosion

Transformers use Self-Attention, which scales at N^2 .

100k points = 10 Billion operations per layer.
Impossible for real-time cars.

Structural Gap

LiDAR points are scattered dots. ViTs were built for structured pixel grids (images).

Standard patches miss the "empty space" inherent in 3D data.

Literature Review: Two Paths

1. 3D Methods (Voxels)

Splits world into 3D cubes. Excellent geometry retention but **computationally expensive** and memory heavy.

2. 2D Projection (Range)

Flattens 3D into a 2D image. **Blazing fast**, but loses 3D spatial relationships during "unrolling."

Innovation: 3 Key Ingredients

01

Convolutional Stem

Translates math-heavy
LiDAR into "image
language."

02

Rectangular Patches

Tokens shaped like 2×8
strips to match LiDAR beam
patterns.

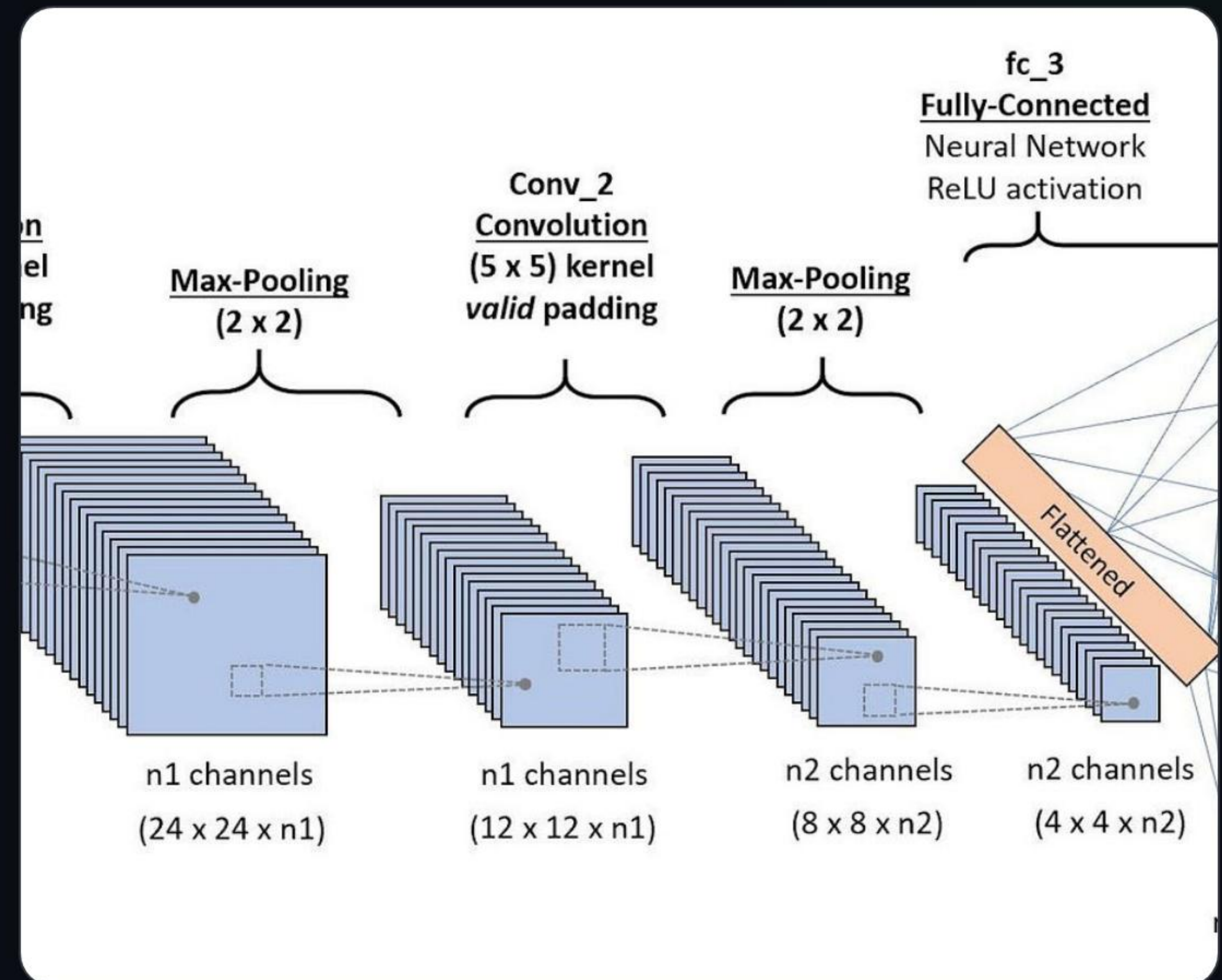
03

Image Pre-training

Borrowing "City Logic"
from billions of labeled
internet photos.

Why the Conv Stem Matters

- Standard ViTs use linear layers to make tokens—this is "too simple" for messy LiDAR.
- **The Solution:** A non-linear stem using SalsaNext blocks.
- It extracts **local context** (edges, surfaces) before the Transformer handles the global scene.



| Knowledge Transfer from Images

+2.8

mIoU Increase

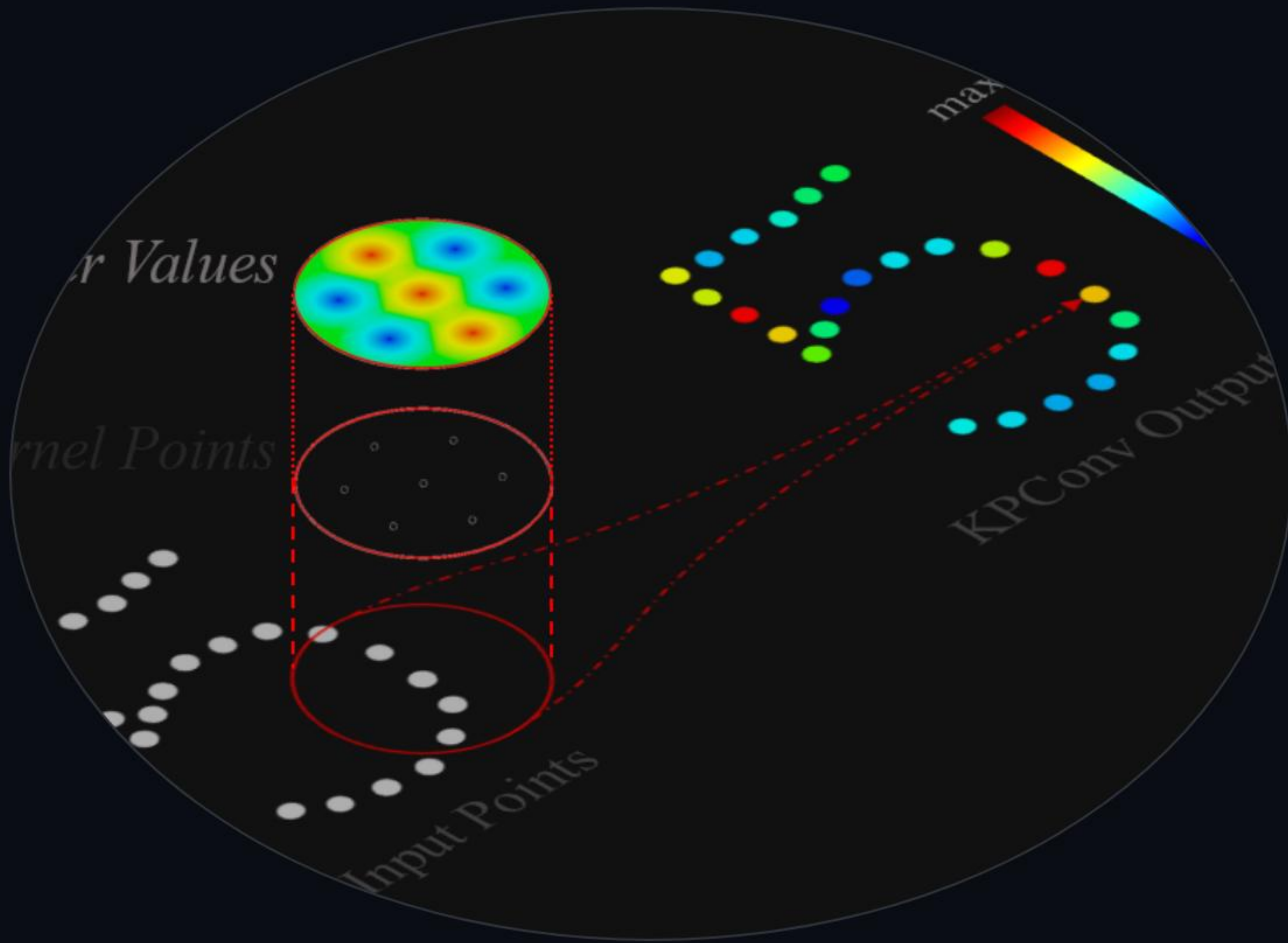
vs. Training from Scratch

Borrowing the "Image Brain"

Labeling 3D data is 10x more expensive than photos.
RangeViT uses a ViT pre-trained on:

- > **ImageNet:** General shapes and objects.
- > **Cityscapes:** Urban layout common sense (Roads are flat, Trees are vertical).

Restoring 3D Geometry



Refinement with KPCConv

2D projection "smudges" object borders (e.g., a tree leaf touching a car in 2D).

The **3D Refiner** projects 2D features back to (x, y, z) points and runs a geometric check to fix depth errors.

Ablation: Proving the Ingredients

Configuration	nuScenes mIoU	Impact
Standard ViT (Baseline)	65.52	-
+ Convolutional Stem	69.82	+4.30
+ UpConv Decoder	73.83	+4.01
+ 3D KPConv Refiner	74.60	+0.77

The Road Ahead

- > **Adaptive Tokens:** Using FlexiViT to change resolution based on distance.
- > **Sensor Fusion:** Combining LiDAR distance with Camera color.
- > **Scaling:** Moving from "Small" to "Large" ViT models as compute improves.

LIDAR units

Creates a 360-degree 3D map of the vehicle's surrounding using lasers

Stereo Cameras

Pairs of cameras are used to take 2D picture, and can detect objects

Global Satellite

Provides location data

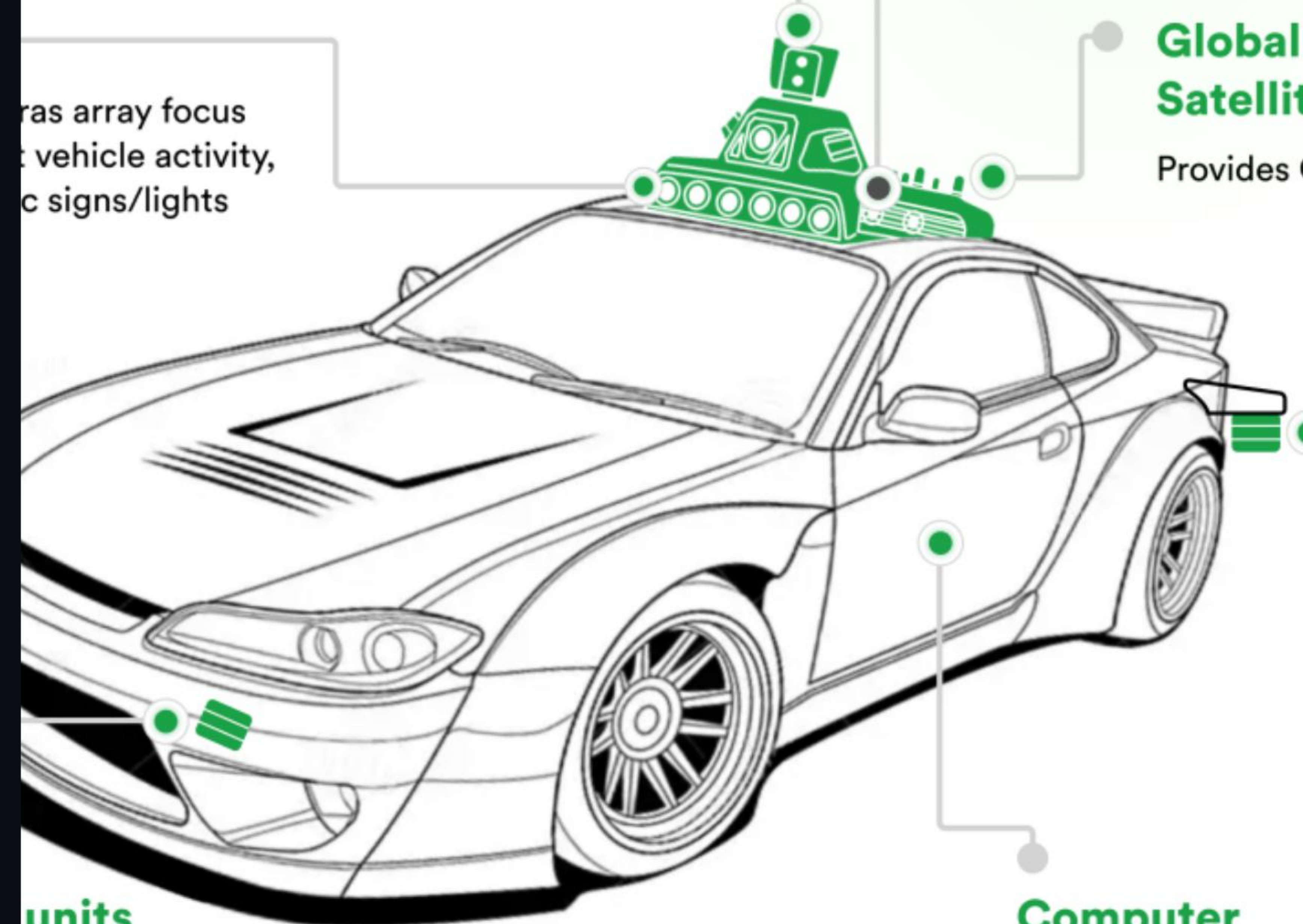
Camera array focus on vehicle activity, traffic signs/lights

Units

Detects objects close to the vehicle

Computer

Built-in computing for data processing



Next Frontier: XAI in LiDAR

Explainable AI (XAI)

Safety is not just about a score; it's about **Transparency**.

Future focus: Visualizing **Attention Maps** to see *exactly* what the car is looking at when it makes a decision.

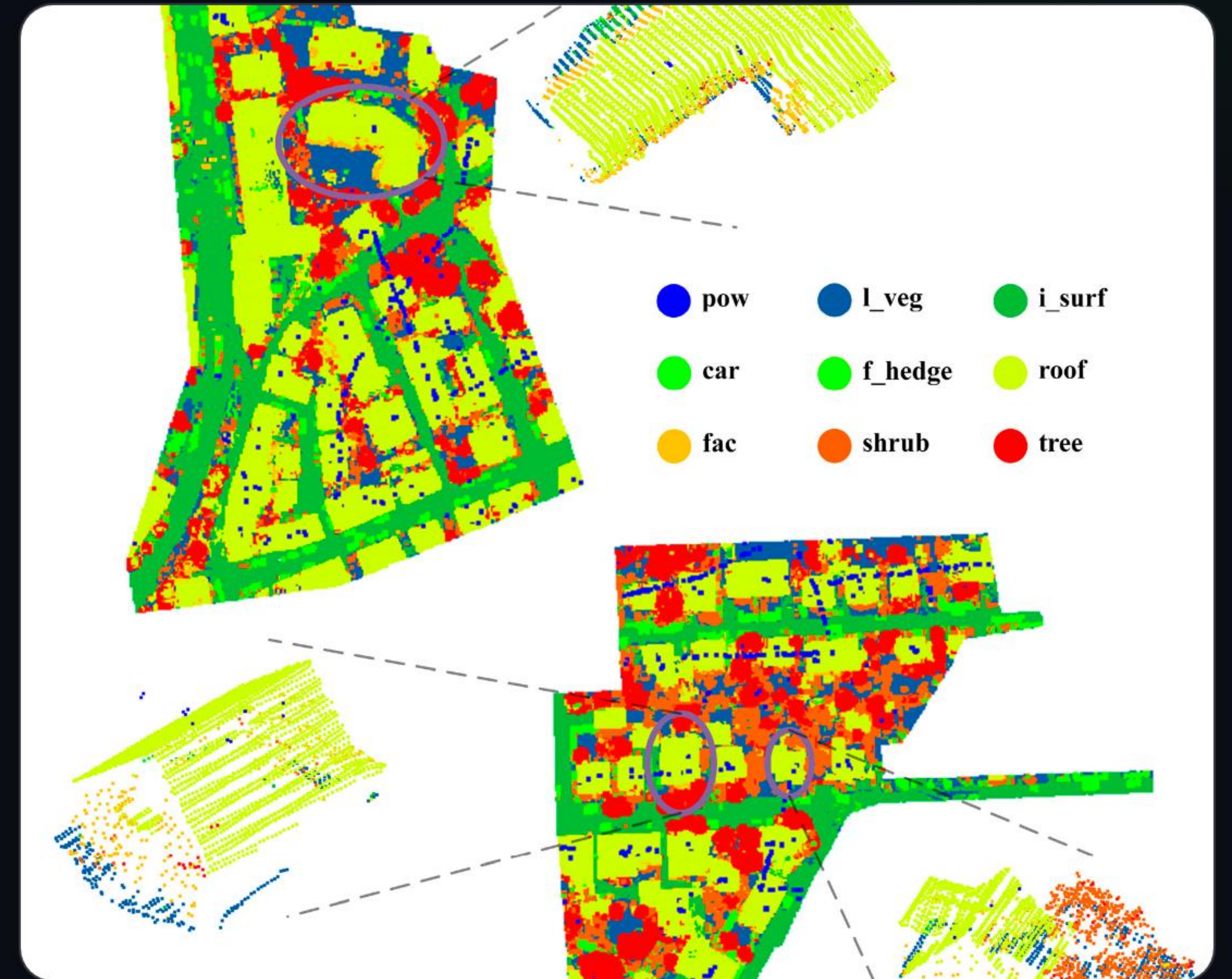
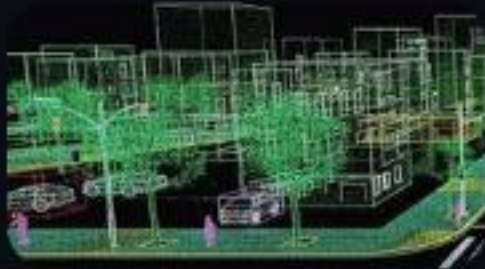
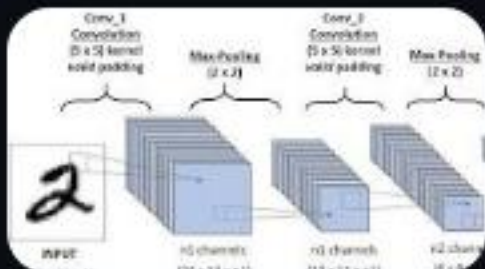


Image Sources



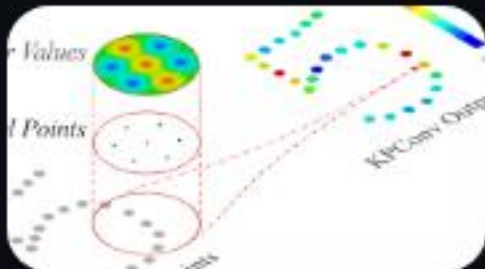
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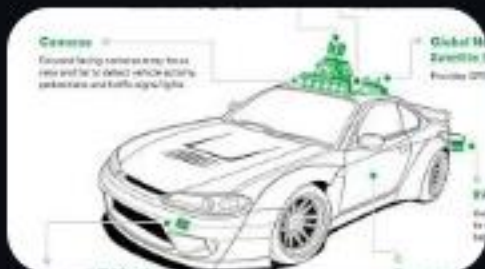
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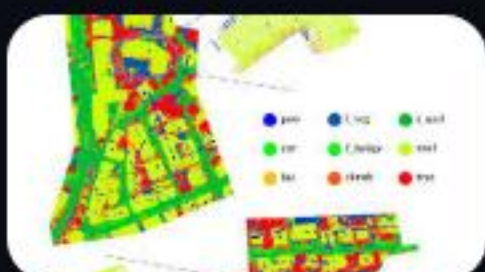
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